

Darah Via Moscoso

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work-portfolio-taupe.vercel.app

EDUCATION

University of the Philippines Visayas <i>Bachelor of Science in Computer Science - cum laude</i>	Miagao, Iloilo 2022 – 2026
Colegio San Agustin - Bacolod <i>Valedictorian</i>	Bacolod City, Negros Occidental 2020 – 2022
Dr. Vicente F. Gustilo Memorial National High School <i>With High Honors</i>	Cadiz City, Negros Occidental 2016 – 2020
Cadiz West Elementary School II <i>Valedictorian</i>	Cadiz City, Negros Occidental 2010 – 2016

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript, Java, C, SQL, TypeScript, PHP, C#, C++, Rust, Dart, Go, Shell Scripting
Software and OS: Microsoft Windows, macOS, Fedora, Kali Linux, Git, Visual Studio Code, Eclipse, Android Studio, Jupyter Notebook, Figma, Conda
Frameworks & Libraries: React, Laravel, Angular, Vue, Next.js, Node.js, Flutter, Bootstrap, TailwindCSS, SCSS/SASS
Database: MySQL, Firebase Firestore, MongoDB, Supabase
Development Methodologies: Agile, Scrum, Software Development Life Cycle (SDLC), Version Control, Test-Driven Development (TDD)
AI Tools: ChatGPT, GitHub Copilot, Google Gemini, Claude
Testing, Validation, and Optimization: Test Case Design, Parameter Tuning and Optimization, Constraint Validation, Functional and Integration Testing, Firmware/Software Debugging, Risk Mitigation and Defect Tracking.
Embedded Systems: ESP32, Arduino, ATtiny85, ATmega328P, AVR Architecture, Communication Protocols (I2C, SPI, UART), Timers and Interrupts, PID Control
Security and Penetration Testing: Penetration Testing, Metasploit, Nikto, Snort, Wireshark, Rootkit Analysis, Botnet, Rogue Wi-Fi Access Point Detection
Network Monitoring: Nagios Core, Proxmox VE, Firewall, SSH, GnuPG

EXPERIENCE

IT Department Intern <i>Department of Education</i>	June 2025 – July 2025 Cadiz City, Negros Occidental
– Served as Lead Developer in a team that developed an Employee Leave and Credit Management System using Laravel. – Developed features for the Non-Teaching Leave module, including employee search, leave application, cancellation, leave credit updates, and leave history. – Implemented the Teaching Leave module to manage leave credits based on policy guidelines and automatically update employee balances. – Built the Compensatory Time-Off (CTO) module to record earned credits, process leave requests, and track available balances. – Added PDF report generation for leave records and transaction history using Laravel DomPDF.	
Teaching and Learning Resource Center (TLRC) Peer Tutor <i>University of the Philippines Visayas</i>	Miagao, Iloilo
– Served as a peer tutor for MATH 53 (Calculus I), MATH 50 (Applied Calculus), and CMSC 123 (Data Structures).	

HACKATHONS JOINED

Third Place, Quantum Computing and Cybersecurity Hackathon <i>Quantum Computing Society of the Philippines</i>	2026 UPV Miagao Campus
First Place, Build-a-Block Hackathon <i>Miagao Valley</i>	2026 UPV Iloilo City Campus

PROJECT EXPERIENCE

Miagao Pickleball Club Queueing System | *Full-Stack Developer*

August 2026

- Next.js, React, JavaScript, State Management, Client-Side Routing, UI/UX Design
- Built a court queueing system used by 100+ active players to manage matches, player rotations, and court assignments during club sessions.
- Developed automatic court timers and match rotations that move finished players back into the queue and assign the next available players.
- Added separate admin and public views, with access controls for managing queues, teams, and court sessions.
- Implemented persistent queue data to keep player positions and active matches intact after page refreshes.
- Tested the system during actual club sessions and adjusted the queue and rotation rules based on how players and organizers used it.

Bloom Daily | *Full-Stack Developer*

September 2026

- Next.js, TypeScript, React, Tailwind CSS, Supabase, PostgreSQL, Server Actions
- Built a full-stack productivity app for managing daily journals, routines, tasks, and focused work sessions.
- Implemented user authentication and protected routes using Supabase Auth and Next.js middleware.
- Designed the database structure and Server Actions for managing daily logs, thoughts, tasks, and customizable routines.
- Added a timetable view, focus timer, habit tracking, and streaks to help users keep track of their daily activities.
- Created reusable and responsive UI components with Tailwind CSS and organized the app using the Next.js App Router.
- Handled loading states, errors, and database operations to keep interactions smooth across the app.

Automated Course Timetabling Using UniTime | *Special Problem Researcher*

May 2026

- Java, UniTime, Constraint Programming, Curriculum-Based Course Timetabling (CB-CTP)
- Configured and evaluated UniTime's constraint solver to automate course scheduling for three degree programs at UP Visayas.
- Modeled academic resources, including courses, instructors, rooms, and curriculum requirements, while implementing scheduling constraints based on department policies and stakeholder requirements.
- Analyzed generated timetables using hard and soft constraint metrics, achieving complete class assignment with no curriculum conflicts and high satisfaction rates for scheduling preferences.
- Performed automated constraint validation and functional testing to verify algorithm adherence to departmental policies and resource availability.
- Documented test metrics, analyzed failure modes for conflicting schedules, and proposed configuration-based mitigation strategies.
- Adjusted solver parameters, search weights, and constraint penalty thresholds to optimize search performance, eliminate curriculum scheduling conflicts, and improve timetable quality

Department of Education School-Based Feeding Program Camera | *Lead Developer*

August 2025

- Dart, Flutter, Provider, Canvas API (`dart:ui`)
- Developed a mobile camera application using Flutter for the Cadiz City Department of Education's School-Based Feeding Program to simplify photo documentation and monitoring.
- Deployed the application for city-wide use, supporting field documentation and compliance monitoring across participating schools.
- Added automatic image overlays, including school information, timestamps, and program logos, before saving photos to the device gallery.
- Implemented image processing using Flutter's Canvas API and managed image storage and cleanup within the application.
- Used the Provider state management package to handle application state and support data for more than 50 district school profiles.

Calendar App (Automated Scheduling System) | *Lead Developer*

December 2023

- Java, Object-Oriented Programming (OOP), Data Structures
- Developed a desktop calendar application with an automated scheduling system based on the Round-Robin algorithm to organize fixed and flexible tasks.
- Used **TreeMap** and **Queue** data structures to manage 15-minute time slots and minimize scheduling conflicts.

- Designed a user-friendly GUI with Java Swing, including input validation and color-coded schedules for improved readability.

PixelPaw Virtual Pet Simulator | *Full-Stack Mobile Developer*

May 2025

- Dart, Flutter, Firebase, Git
- Contributed to the development of a Flutter-based virtual pet simulation game by implementing gameplay features, user interface improvements, and interactive game elements.
- Developed mini-games including Tic-Tac-Toe and Flappy Bird, integrating game logic, scoring, player interactions, and leaderboard features.
- Implemented pet lifecycle features such as death states, pet creation, local data storage, and gameplay condition handling.
- Improved application interface through updated layouts, custom assets, dialogs, animations, sound effects, and responsive button interactions.

ESP32 Sungka Game | *Hardware-Software Developer*

December 2025

- Arduino IDE, ESP32, PCF8575 I/O Expander, I2C, SPI, ILI9341 TFT Display, Embedded Systems
- Contributed in developing a digital version of the traditional Filipino board game Sungka using an ESP32 microcontroller and embedded hardware components.
- Integrated an I2C-based PCF8575 I/O expander for additional button inputs and implemented real-time game display using a TFT screen.
- Designed and executed functional and hardware integration test cases to validate low-level I2C data signals from the PCF8575 expander and SPI communication with the ILI9341 TFT screen.
- Conducted boundary and edge-case testing on embedded state transitions, interrupts, and game logic across C++ modules to ensure long-run firmware stability.
- Contributed in programming game mechanics including turn management, seed distribution, capturing rules, and score tracking.
- Worked with a team to assemble and package the prototype into a functional standalone device.

Multitape Turing Machine Simulator | *Lead Developer*

December 2024

- Python, Object-Oriented Programming (OOP), Turing Machine Theory
- Developed a Python-based simulator for designing and executing multitape Turing Machines.
- Built a modular architecture to separate the machine logic from data structures for easier maintenance and extension.
- Implemented support for multiple tape heads, state transitions, and simultaneous read/write operations across tapes.

REFERENCES

MA. THERESA AGNES D. MARCELLA

Administrative Officer V

Office of the Schools Division Superintendent (OSDS), Department of Education Cadiz City

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ASST. PROF. FRANCIS D. DIMZON

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